DOMailer

Next-Gen Email Workspace

A. Case Study

You've just joined **TechCorp**, a fast-growing tech startup, as a **Frontend Engineer**. Your team is building **DOMailer**, an internal workspace email client designed to boost employee productivity.

The UI/UX team has already handed over the static visual designs. Your main task now is to bring this interface to life using **Vanilla JavaScript** by leveraging **DOM (Document Object Model) Manipulation**.

DOMailer must not be a simple read-only email client. It requires a fast, responsive, and interactive email management system featuring automated tagging, real-time filtering, live data calculation, and bulk action selection.

B. Learning Objectives

Through this assignment, you're expected to demonstrate proficiency in:

1. Transforming data structures (Array/Object) into dynamic visual elements in the DOM.
2. Implementing **Event Delegation** for performant and scalable event handling.
3. Navigating node hierarchies (parentElement, closest, children).
4. Managing component states using CSS classes (classList) and data attributes (dataset).
5. Handling forms, user inputs, and updating live UI counters in real time.

C. Feature Requirements

Feature 1: Dynamic Email Rendering & Live Counters

- Dynamically render the email list from the provided mock array into the #email-list container.

- Display live metrics for **Total Unread Emails** and **Total Starred Emails** in the sidebar/header.
- Counters must **automatically recalculate** whenever any email status changes (read, starred, deleted, or added).

Feature 2: Advanced Action Panel (Single & Bulk Selection)

- **Checkbox Selection:** Include a selection checkbox on each email row.
- **Select All:** Provide a master checkbox at the top to select or deselect all visible emails at once.
- **Bulk Action Bar:** When at least one email is checked, display a contextual toolbar allowing users to:
    - *Mark as Read:* Update the status of all checked emails to read.
    - *Delete Selected:* Remove all checked email nodes from the DOM simultaneously.

Feature 3: Interactive Row Actions (Event Delegation Mandatory)

*Strict Constraint: Attaching individual event listeners to each email row is strictly prohibited. You must use **Event Delegation** on the parent container (#email-list).*

- **Toggle Star:** Click the star icon to mark an email as important (toggle active star visual state).
- **Mark as Read/Unread:** Click an email row to toggle its read status (change typography from bold to normal).
- **Delete Single Email:** Click the delete icon to remove that specific row directly from the DOM.

Feature 4: Real-Time Smart Filter & Search

- Implement a search input. As the user types, filter the visible email list by sender name or subject line in real time (input event).
- Provide category filter buttons: **All**, **Unread**, and **Starred** to narrow down the view.

Feature 5: Quick Compose with Auto-Categorization

- Create a "Compose Email" modal form or slide-out panel.
- Upon form submission:
    - Prevent default page reload (e.preventDefault()).
    - Construct a new email element and insert it at the top of the email list (prepend).

- *Challenge Feature:* If the subject line contains the word "URGENT" or "IMPORTANT", automatically apply a high-priority red tag to that email row.

Layout Example

